3.2 Software Architecture

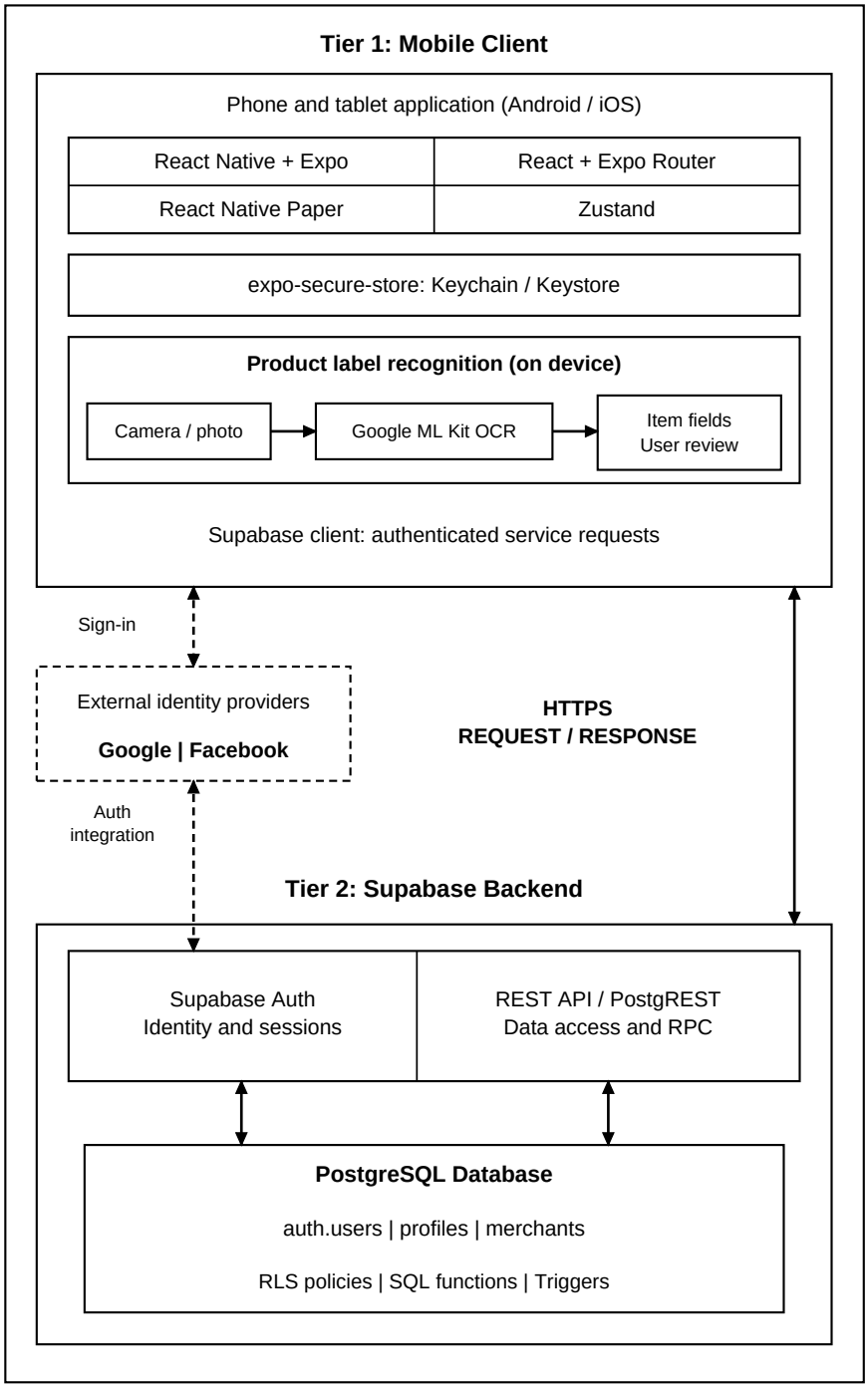


Figure 3.1.1: 2-Tier Software Architecture Diagram

This illustrates the 2-Tier Software Architecture of the Mobile Merchant System. The architecture consists of the Mobile Client Tier and the Supabase Backend Tier. The application communicates directly with managed backend services through HTTPS, without a separate custom application server. Authentication, API access, and database storage are shown as

components within the backend tier. Google and Facebook are external identity providers and are not counted as additional application tiers.

The Mobile Client Tier provides the interface and application interactions on Android and iOS phones and tablets. React Native and Expo support the native application, while React, Expo Router, and React Native Paper provide screen composition, navigation, and interface components. Zustand manages session state, and expo-secure-store retains session information using the device Keychain or Keystore. Google sign-in uses the native sign-in interface, while Facebook authentication uses the system browser and returns to the application through its registered deep link.

Google ML Kit OCR operates within the Mobile Client Tier to recognize text from product-label photographs captured by the camera or selected from the device. The recognized text is used to populate applicable item fields for user review before saving. Text recognition runs locally on the phone and does not require a remote OCR server. This component therefore remains part of the client tier, while saving records to Supabase requires communication with the backend.

The Supabase Backend Tier provides authentication, data access, and persistent storage. Supabase Auth verifies sign-in credentials and manages authenticated sessions. The REST API, implemented through PostgREST, exposes permitted data operations and database function calls to the application. PostgreSQL stores authentication records, user profiles, and merchant records. Row-Level Security policies enforce access restrictions, while SQL functions and triggers support profile creation, merchant access checks, and account deletion. The mobile client sends authenticated requests to the managed services, which interact with the database and return results. This arrangement separates local interaction and recognition from centralized identity, authorization, and data management.